

Ushtrime ND5c

1.3.34 Guri i rëndë $Q=20\text{ N}$ bie nga lartësia $h=5\text{ m}$ dhe futet $\Delta l=5\text{ cm}$ në tokë të butë. Sa është forca mesatare e goditjes në tokë?

Zgjidhja

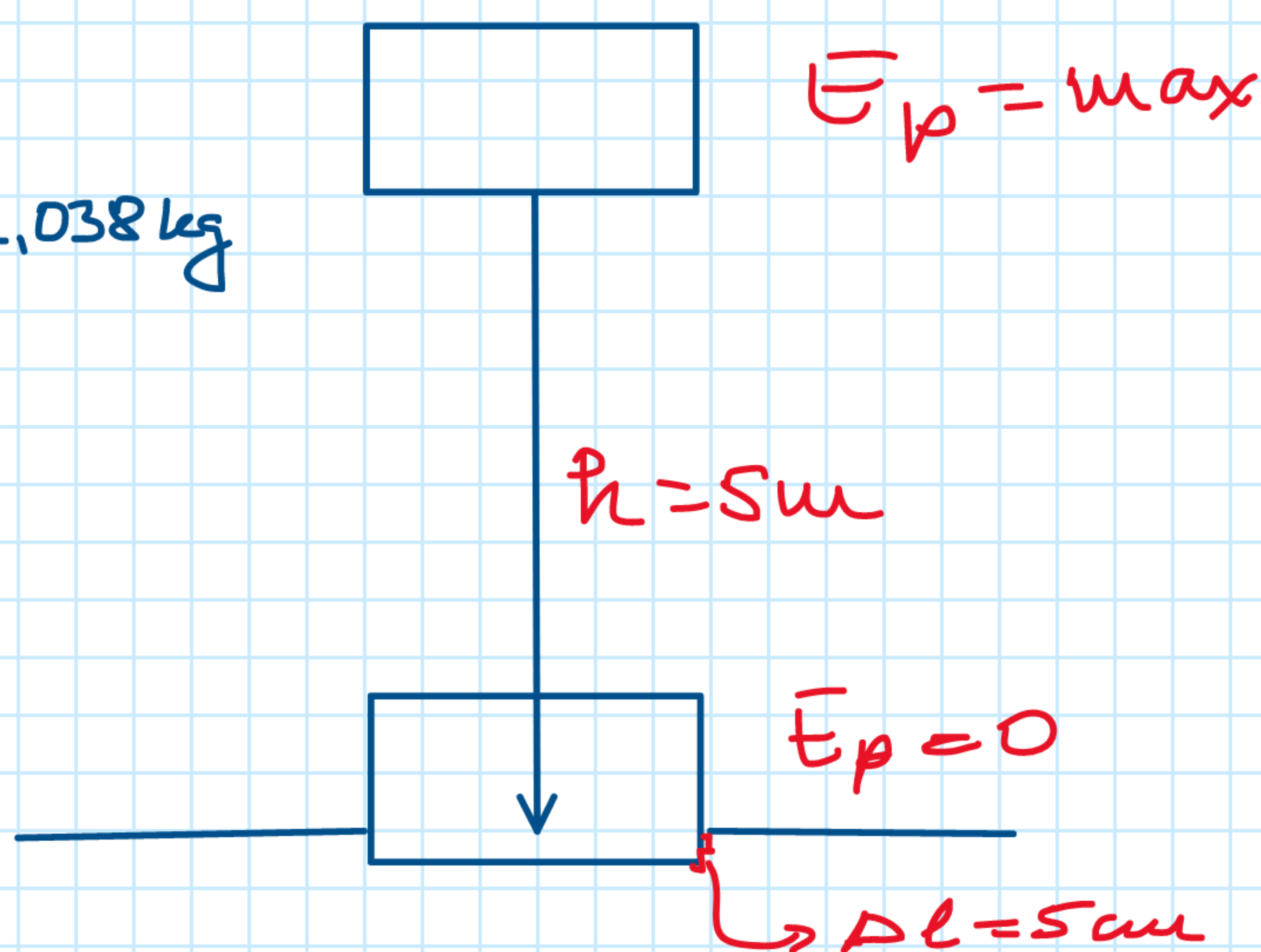
Të dhënat

$$Q=20\text{ N} \quad m = \frac{20\text{ N}}{9,81\frac{\text{m}}{\text{s}^2}} = 2,038\text{ kg}$$

$$h=5\text{ m}$$

$$\Delta l = 5\text{ cm} = 5 \cdot 10^{-2}\text{ m}$$

$$F=?$$



$$E_p = A \rightarrow Q \cdot h = F \cdot \Delta l / : \Delta l$$

$$A = F \cdot \Delta l$$

$$E_p = Q \cdot h$$

$$F = \frac{Q \cdot h}{\Delta l} = \frac{20\text{ N} \cdot 5\text{ m}}{5 \cdot 10^{-2}\text{ m}} = 2010^2\text{ N}$$

$$F = 2 \cdot 10^3\text{ N} = \underline{2\text{ kN}}$$

1.3.35 Ujëvara e lartë $h=5\text{ m}$ në çdo sekondë jep $V=15\text{ m}^3$ ujë. Ç'fuqi ka ujëvara?

Zgjidhja

Të dhënat:

$$h=5\text{ m} ; t=1\text{ s}$$

$$V=15\text{ m}^3 (\text{H}_2\text{O})$$

$$P=?$$

$$\rho = 1000\frac{\text{kg}}{\text{m}^3}$$

$$t = 4^\circ\text{C}$$

$$s = h$$

$$P = \frac{A}{t} = \frac{F \cdot s}{t} = \frac{mg \cdot s}{t}$$

$$P = \frac{mgh}{t} = \frac{F_p}{t}$$



$$P = \frac{mgh}{t} = \frac{\rho \cdot V \cdot gh}{t} = \frac{1000\frac{\text{kg}}{\text{m}^3} \cdot 15\text{ m}^3 \cdot 9,81\frac{\text{m}}{\text{s}^2} \cdot 5\text{ m}}{1\text{ s}}$$

$$\frac{m}{V} = \rho \rightarrow m = \rho \cdot V$$

$$P = 735750 \frac{\text{kg} \cdot \frac{\text{m}}{\text{s}^2} \cdot \text{m}}{\text{s}}$$

$$= 735,75 \cdot 10^3 \frac{\text{N} \cdot \text{m}}{\text{s}} = 735,75\text{ kW}$$

$$\frac{\text{kg} \cdot \frac{\text{m}}{\text{s}^2} \cdot \text{m}}{\text{s}} = \frac{\text{N} \cdot \text{m}}{\text{s}} = \frac{\text{J}}{\text{s}} = \text{W}$$